Instrumentation & Controls Engineer

SSOE Group · Austin, TX

Job ID (this archive)	MFG-060
Employer	SSOE Group
Location	Austin, TX
Career stage	Experienced
Location group	Texas (outside DFW)
Major (archive tag)	Electrical Engineering
Sector area	Electronics & semiconductor manufacturing
Industry (LinkedIn)	Engineering Services
Pay listed	$80,000 - $130,000/year
Stated rules	No GPA or sponsorship rule stated
Posted / closes	2026-10-29 · 2027-01-09
Status when checked	Accepting applications on LinkedIn (Nov 1, 2026)
Source page	https://www.linkedin.com/jobs/view/4440326216/
Captured	2026-11-01

Full posting text as it appeared on LinkedIn on Nov 1, 2026, saved for ECS 1100. Postings change and close often; open the source page above for the current version before applying.

At a glance

What you would do

Engineer designing instrumentation and control systems for semiconductor cleanrooms, producing AutoCAD/Revit drawings, sequences of operation and supporting construction and commissioning.

Who can apply

- **Majors named:** Instrumentation Engineering, Electronics and Instrumentation, Control Systems, Electrical Engineering
- **Degree and experience:** Bachelor's in Instrumentation, Electronics & Instrumentation, Control Systems, Electrical Engineering or related; 5+ years I&C design experience

Skills mentioned

Teamwork & collaboration · Communication · Automation & controls · Safety · Excel · CAD · Cybersecurity & OT security

Full posting text

Within SSOE’s Tool Install division, we specialize in high-tech Tool Installation design, offering top-tier design and engineering services to a diverse array of clients and markets. Our team is composed of highly skilled and knowledgeable professionals who excel across various disciplines. We pride ourselves on our cross-disciplinary approach, which allows us to understand and address our clients' needs more comprehensively than traditional, siloed methods. What sets us apart is our ability to seamlessly conceptualize, plan, and detail every aspect of design. Our in-house experts collaborate closely, optimizing and enhancing our work through multi-discipline strategy sessions. By leveraging cutting-edge techniques and state-of-the-art software, we deliver industry-leading technology and innovative solutions from day one.

We are seeking an Instrumentation and Controls Engineer to join our Tool Install team. In this position, you will lead the design and delivery for complex semiconductor and high-tech manufacturing facilities, mentor the next generation of engineers, and drive projects to completion. This role designs control systems, instrumentation selections, sequences of

operation, and coordinated construction documents using AutoCAD and Revit, while working closely with project managers, discipline leads, clients, and contractors. The engineer will review design changes, manage technical risks, support QA and QC checks, and help ensure constructability, safety, and operational reliability from early design through commissioning and startup. Success in this role requires strong technical judgment, clear communication, and a collaborative approach in a fast-paced global project environment.

AI and Innovation

At SSOE, we don’t just talk about the future, we build it. As a leader in industry AI adoption, we empower our team to work smarter by providing responsible access to the industry leading AI Large Language Models, vibe-coding platforms, and securely integrating advanced tools into every role. From day one, you’ll receive the training and support needed to leverage AI as a core tool for enhancing productivity, sparking creativity, and streamlining complex workflows.

Responsibilities

- Develop instrumentation and controls design solutions for cleanrooms, fab facilities, and advanced manufacturing support systems across conceptual, schematic, design development, and construction document phases.
- Prepare design calculations, equipment and device schedules, control narratives, sequences of operation, I/O lists, and related technical documentation for building and process support systems.
- Create and coordinate drawings and models in AutoCAD and Revit, including plans, details, sections, risers, and coordinated layouts for control devices, panels, and associated infrastructure.
- Support integration of controls for HVAC, cleanroom pressure management, exhaust systems, process utilities, alarms, monitoring systems, and other facility automation interfaces.
- Coordinate with architectural, mechanical, electrical, process, and fire protection teams to resolve spatial conflicts, interface requirements, and system dependencies early in the design process.
- Participate in internal reviews, QA and QC checks, and interdisciplinary coordination meetings to ensure technical accuracy, code compliance, and alignment with project intent.
- Review vendor submittals, product data, shop drawings, and equipment information to confirm compliance with design requirements and project specifications.
- Respond to RFIs, review proposed field changes, and support formal change control by assessing technical impact, constructability, cost, and schedule implications.
- Conduct site walks, observe installation progress, verify field conditions, and support punch list resolution, testing, and commissioning activities during construction administration and startup.
- Coordinate directly with clients, contractors, and vendors to clarify requirements, resolve issues, and maintain project momentum through timely communication and follow-up.
- Identify design risks, safety concerns, and operational constraints, and recommend practical solutions that support reliable and maintainable facility performance.
- Mentor junior team members by reviewing drawings, sharing technical guidance, and supporting consistent design practices across project teams.

Qualifications

- Bachelor’s degree in Instrumentation Engineering, Electronics and Instrumentation, Control Systems, Electrical Engineering, or a closely related field.
- Minimum of 5+ years of experience in instrumentation and controls design for industrial, high tech, semiconductor, pharmaceutical, life sciences, or similarly complex facilities.
- Strong working knowledge of Tool Install cleanroom and fab support systems, including monitoring and control of HVAC, pressure relationships, temperature, humidity, airflow, alarms, and process utility interfaces.
- Proficiency with AutoCAD and Revit for preparing and coordinating engineering drawings and models; experience with other controls design or documentation tools is a plus.
- Ability to prepare accurate control narratives, sequences of operation, I/O lists, design calculations, and coordinated construction packages.
- Experience supporting design reviews, submittal reviews, RFIs, field coordination, commissioning, and startup activities on deadline driven projects.

- Strong written and verbal communication skills, with the ability to work effectively with stakeholders, contractors, clients, and multidisciplinary teams.
- Demonstrated ownership, attention to detail, and a collaborative mindset.
- Ability to mentor junior staff, manage multiple priorities, and contribute to quality focused project execution in a deadline driven environment.
- Salary Range: $80,000 - $130,000/year depending on location, education, depth of relevant experience, and any specialized licensure/certifications. Opportunity for ample rewards, leadership incentive components, and generous bonus opportunities.
- Benefits: Includes comprehensive health, dental, and vision insurance, life insurance, 401K retirement savings plan (with company matching), opportunities for professional development and training (including support for relevant licensure & certifications), and generous PTO & paid holidays.

Join us at SSOE, where your expertise will thrive in a dynamic, collaborative environment dedicated to pushing the boundaries of design and engineering excellence. Our company culture is built on collaboration, innovation, and a commitment to excellence. We believe in fostering a dynamic and inclusive environment where every team member can thrive. If you're passionate about innovation and eager to make a significant impact, we invite you to apply and become a part of our forward-thinking team.

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